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Credit Freezes, Equilibrium Multiplicity, and Optimal Bailouts in Financial Networks

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1 Big picture

- **Question.** How can financial-network interdependencies generate self-fulfilling insolvencies and credit freezes, and what is the cheapest way for a regulator to prevent them?
- **Core mechanism.** A bank's value depends on payments from other banks. If one bank defaults or delays payment, its creditors may default or delay payments too. When these dependencies feed back through a cycle, pessimistic beliefs can become self-fulfilling: everyone could pay in a good equilibrium, but no one pays in a bad equilibrium.
- **Main theoretical result.** Equilibrium multiplicity is a network-cycle phenomenon. If there is no relevant dependency cycle, the best and worst equilibrium values coincide. If there is a dependency cycle, then for some portfolio values and failure costs, the network can have multiple equilibria.
- **Systemic-solvency result.** In the best equilibrium, all banks are solvent exactly when the network is weakly balanced: each bank can meet liabilities if incoming claims are fully paid. In the worst equilibrium, weak balance is not enough. There must also be an iteratively strongly solvent set that intersects every directed simple cycle.
- **Bailout problem.** The regulator chooses capital injections before bankruptcy costs or payment delays occur. Minimum bailouts in the best equilibrium are simple: inject each bank's net imbalance. Minimum bailouts that rule out self-fulfilling insolvencies in nonbest equilibria are much harder because the order of bailouts matters.
- **Computational result.** Finding the minimum-cost bailout policy for nonbest equilibria is strongly NP-hard. The difficulty comes from overlapping cycles: bailing out one bank changes the bailout cost and indirect bailout value of many others.
- **Policy insight.** Indirect bailouts matter. Instead of injecting capital directly into a distressed bank, it can be cheaper to inject capital into its debtors, causing them to repay it and thereby restoring the target bank indirectly.
- **Quantitative/network insight.** A greedy policy based on indirect bailout value over bailout cost is not always optimal, but it guarantees systemic solvency at a cost no larger than one-half of the relevant overall shortfall. In star and core-periphery networks, it is often optimal to begin with peripheral banks rather than large core banks.
- **Economic takeaway.** Financial fragility is not only about contagion from bad fundamentals. It is also about equilibrium selection. Cycles of obligations can create credit freezes, and well-designed interventions exploit the network rather than simply rescuing the largest institutions first.

2 Model and primitives (key objects)

2.1 Financial network

- The network contains nodes $N = \{0, 1, \dots, n\}$.
 - Banks are nodes $1, \dots, n$.
 - Node 0 aggregates outside actors that are not themselves financial-network dominoes.

- Bank i has outside assets with realized value p_i . The vector is $p = (p_i)_i$.
- D_{ij} is the face value of debt owed by debtor j to creditor i . Hence

$$D_i^A \equiv \sum_j D_{ij}, \quad D_i^L \equiv \sum_j D_{ji},$$

where D_i^A is bank i 's nominal interbank assets and D_i^L is bank i 's nominal interbank liabilities.

- Edges point from debtor to creditor. A directed cycle is therefore a cycle of payment obligations. Such cycles are the key source of self-fulfilling defaults.

Interpretation. The paper takes the network and outside-asset realizations as given. The relevant moment is crisis settlement: banks cannot quickly redesign their liabilities, so solvency depends on who pays whom.

2.2 Default, repayment, and failure costs

Let $d_{ij}(V)$ be the actual payment from debtor j to creditor i , as a function of the vector of bank values V .

If debtor j is solvent, it pays in full:

$$d_{ij}(V) = D_{ij}.$$

If debtor j defaults, creditors share its remaining asset value pro rata:

$$d_{ij}(V) = \frac{D_{ij}}{\sum_h D_{hj}} \max \left\{ p_j + d_j^A(V) - \beta_j(V, p), 0 \right\}, \quad (1)$$

where $d_j^A(V) = \sum_i d_{ji}(V)$ is the realized value of bank j 's incoming payments.

Failure costs are imposed only when a bank is insolvent:

$$b_i(V, p) = \begin{cases} 0, & \text{if } p_i + d_i^A(V) \geq D_i^L, \\ \beta_i(V, p), & \text{if } p_i + d_i^A(V) < D_i^L. \end{cases} \quad (2)$$

A useful canonical case is

$$\beta_i(V, p) = b + a \left(p_i + d_i^A(V) \right), \quad b \geq 0, \quad a \in [0, 1]. \quad (3)$$

Interpretation. Failure costs are the discontinuous loss from bankruptcy, fire sales, delayed payments, legal costs, or operational disruption. They reduce what creditors receive, so they are exactly what makes a payment freeze costly and potentially self-fulfilling.

2.3 Equilibrium values

Bank i 's book value is

$$V_i = p_i + d_i^A(V) - D_i^L - b_i(V, p). \quad (4)$$

In vector notation:

$$V = p + d^A(V) - D^L - b(V, p). \quad (5)$$

A vector V is an equilibrium if it solves this fixed-point equation.

Key point. Because one bank's repayment depends on other banks' repayment, Equation (5) can have multiple fixed points. Monotonicity implies a lattice of equilibria: a best equilibrium, a worst equilibrium, and possibly intermediate equilibria.

3 Solvency concepts (definitions used throughout)

3.1 Balance conditions

- **Weak balance:**

$$p_i + D_i^A \geq D_i^L.$$

Bank i is solvent if all its incoming claims are paid in full.

- **Exact balance:**

$$p_i + D_i^A = D_i^L.$$

The bank has no buffer when incoming claims are fully valued.

- **Critical balance:**

$$p_i + D_i^A - D_{ij} < D_i^L \quad \text{for every } j \text{ with } D_{ij} > 0,$$

plus weak balance. Missing any one incoming claim makes the bank insolvent.

Interpretation. Weak balance is enough for the optimistic clearing outcome. Critical balance captures fragility: even one unpaid counterparty can trigger default.

3.2 Unilateral and iterative solvency

A bank is **unilaterally solvent** if

$$p_i \geq D_i^L.$$

It can pay even if nobody pays it.

A set S is **iteratively strongly solvent** if it can be written as $S = S_1 \cup \dots \cup S_K$ such that:

- banks in S_1 are unilaterally solvent;
- for each $k \geq 2$, banks in S_k become solvent after receiving payments from previously solvent sets:

$$p_i + \sum_{j \in S_1 \cup \dots \cup S_{k-1}} D_{ij} \geq D_i^L.$$

Interpretation. Iterative solvency is a repayment cascade. Some banks can pay without help; their payments make more banks solvent; those banks' payments make still more banks solvent.

4 Main theoretical results (all equations + interpretation)

4.1 Equilibrium multiplicity

Proposition 1: cycles and multiplicity.

- If there is no dependency cycle, then the worst and best equilibria coincide.
- If there is a dependency cycle, then there exist portfolio values and failure costs that generate multiple equilibria.
- Every nonbest equilibrium can be understood as the best equilibrium plus additional self-fulfilling default cycles and their outward default paths.

Plain English. Acyclic networks settle recursively: start from banks that depend on nobody, then move outward. Cyclic networks can feed pessimism back to its origin. That feedback is what makes a credit freeze possible.

4.2 Systemic solvency in best and worst equilibria

Proposition 2: characterization of systemic solvency.

- All banks are solvent in the **best equilibrium** if and only if the network is weakly balanced.
- All banks are solvent in the **worst equilibrium** if and only if:
 1. the network is weakly balanced; and
 2. there exists an iteratively strongly solvent set that intersects every directed simple cycle.

Plain English. In the best equilibrium, all claims are effectively trusted, so weak balance is enough. In the worst equilibrium, cycles can freeze unless each cycle is touched by a seed of solvency that can start a repayment cascade.

4.3 Why this is stronger than standard contagion

Standard contagion asks: after a bad fundamental shock, how far do defaults spread? This paper adds a different question: can the same fundamentals support both a good and a bad outcome?

- In the best equilibrium, a weakly balanced cycle can clear.
- In the worst equilibrium, the same cycle can fail to clear if no sufficiently strong bank starts payments.
- Thus the issue is not only fundamentals; it is also coordination over payments.

5 Minimum bailouts (policy problem)

5.1 Regulator's problem

The regulator can inject capital $t_i \geq 0$ into bank i before failure costs are incurred. The minimum bailout problem is

$$\min_{(t_1, \dots, t_n) \in \mathbb{R}_+^n} \sum_i t_i \quad \text{s.t.} \quad V(p + t) \geq 0. \quad (\text{OPT})$$

Important detail. Because there can be multiple equilibria, the constraint must specify which equilibrium is selected. Preventing defaults in the best equilibrium is a different problem from preventing self-fulfilling defaults in the worst equilibrium.

5.2 Best-equilibrium bailouts are simple

For the best equilibrium, the minimum transfer to bank i is its net imbalance:

$$t_i = [D_i^L - D_i^A - p_i]^+.$$

Interpretation. If a bank is not weakly balanced, it needs enough capital to become weakly balanced. Network topology does not matter much here because the best equilibrium assumes all solvent payment chains clear.

5.3 Worst-equilibrium bailouts are hard

For nonbest equilibria, after correcting net imbalances, the regulator must inject enough capital to create an iteratively strongly solvent set that intersects all defaulting cycles.

Why order matters.

- Bailing out bank i can make i pay its creditors.
- Those creditors may then become solvent and pay their own creditors.
- Therefore the cost of bailing out bank j depends on whether i was bailed out first.

Proposition 3. Finding whether there exists a bailout policy with cost below a budget is strongly NP-hard. Therefore, solving (OPT) is also strongly NP-hard.

Plain English. The regulator cannot generally solve the problem by checking banks one by one. Overlapping cycles create a combinatorial problem in which the relevant object is an ordered bailout sequence.

5.4 Indirect bailout value and an upper bound

The paper defines a bank's indirect bailout value: if bank i is rescued, its repayments reduce the shortfalls of its creditors. A first-step version is approximately

$$\sum_j \min \left\{ D_{ji}, \left(D_j^L - p_j - \sum_{k \in S} D_{jk} \right)^+ \right\},$$

where S is the set of already solvent banks. The bailout cost of i is

$$\left(D_i^L - p_i - \sum_{j \in S} D_{ij} \right)^+.$$

A natural greedy policy bails out the bank with the highest ratio

$$\frac{\text{indirect bailout value}}{\text{bailout cost}}.$$

Proposition 4. For any $1 \leq k \leq n - 1$, bailing out banks in decreasing order of k -step indirect bailout value over bailout cost until all are solvent costs at most

$$\frac{1}{2} \sum_i (D_i^L - p_i)^+.$$

Interpretation. The greedy policy is not generally optimal, but it gives a strong systemwide bound: the regulator can always prevent self-fulfilling default at no more than half of total shortfall.

6 Network structures (where intuition becomes sharp)

6.1 Disjoint cycles

If each bank lies on at most one cycle, cycles do not interfere much with one another. The optimal policy is to move through cycles in an appropriate order and bail out the bank closest to solvency on each cycle, accounting for payments induced by earlier bailouts.

Proposition 5. If each bank lies on at most one cycle, the optimal bailout policy consists of bailing out the bank closest to solvency on each simple cycle, accounting for previous repayment cascades.

6.2 Star networks

A star has one center bank and many peripheral banks. Peripheral banks owe the center, and the center owes the peripheral banks. Let D^{in} be the periphery-to-center obligation and D^{out} the center-to-periphery obligation.

Define

$$m^* = \frac{(n-1)D^{out} - p_n}{D^{in}}.$$

The policy first bails out the peripheral banks with the highest portfolio values. Then the regulator compares:

$$(n-1)D^{out} - p_n - \lfloor m^* \rfloor D^{in} \quad \text{versus} \quad D^{in} - p_{\lfloor m^* \rfloor + 1}.$$

Proposition 6. It is optimal either to bail out the first $\lfloor m^* \rfloor + 1$ peripheral banks, or to bail out $\lfloor m^* \rfloor$ peripheral banks and inject the remaining shortfall into the center.

Economic intuition. The center is large, but rescuing it directly can waste the outside assets of the periphery. Rescuing peripheral debtors first lets the regulator leverage their balance sheets.

6.3 Core-periphery networks

In a core-periphery network, core banks are densely linked to each other and each has attached peripheral banks. The star logic still applies locally: to restore a core bank, it is often cheapest to rescue its peripheral debtors first.

Proposition 7. Under the paper's symmetry and integer conditions, an optimal policy first targets the relevant peripheral banks of a core bank before injecting directly into the core bank.

Important caveat. The periphery-first intuition is robust, but once the core itself is asymmetric, the order in which core banks should be restored can again be computationally complex.

7 Simulation exercise (how policies compare)

7.1 Random core-periphery networks

The simulation fixes a core-periphery structure:

- n_C core banks are exposed to each other;
- each core bank has n_P peripheral banks;
- exposures and core-bank outside assets are drawn from uniform distributions;
- peripheral banks have common outside assets p_P ;
- a parameter α controls asymmetry among core banks.

The paper compares four bailout policies:

- **Core First, by Size:** rescue larger core banks first.
- **Core First, by IBV/BC:** rescue core banks by indirect bailout value over bailout cost.
- **Periphery First:** before rescuing a core bank directly, check whether rescuing its periphery is cheaper.
- **Optimum:** compute the minimum-cost policy for the simulated network.

7.2 Simulation takeaways

- Periphery First improves substantially over Core First when peripheral banks have intermediate outside assets.
- If peripheral banks are too weak, there is little outside capital to leverage; if they are already strong, bailouts are less needed. The largest gains occur in the middle.
- With partial failure costs, total bailout costs fall, but the indirect-bailout logic remains.
- When core banks are symmetric, Periphery First is close to optimal or optimal. When core banks are highly asymmetric, the remaining gap to the optimum grows because ordering the core-bank rescues becomes more difficult.

8 Assumptions and interpretation (what makes the theory work)

- **Fixed network.** Exposures are taken as given. The model is most relevant in a crisis, when banks cannot rapidly redesign their liabilities.
- **Failure costs.** Multiplicity relies on a discontinuous cost of default or payment delay. Without such costs, the standard clearing logic is less prone to self-fulfilling insolvency.
- **Equilibrium selection.** The paper distinguishes best and worst equilibria. Bailout needs depend on which equilibrium the regulator wants to rule out.

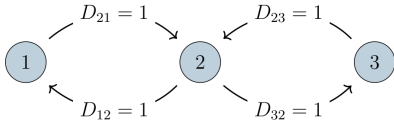
- **Full failure costs in some results.** Several clean characterizations use the benchmark in which a defaulting bank pays nothing until it is solvent. This captures payment freezes and delayed settlement. Some complexity and intuition results are more general.
- **No ex ante moral hazard analysis.** The paper studies ex post crisis intervention. It does not claim regulators should always bail out banks; it asks how to do so at minimum cost conditional on intervening.
- **Regulator information.** The optimal policy requires detailed knowledge of exposures and outside assets. This motivates better financial-network data and netting/compression infrastructure.

9 Figures and tables (extracted visuals)

The paper contains figures but no main tables. The visuals below are extracted from the paper and displayed without figure/table numbering; each visual keeps its substantive title. Adjacent visuals are placed side by side.

9.1 Equilibrium multiplicity and indirect cycles

An example of equilibrium multiplicity



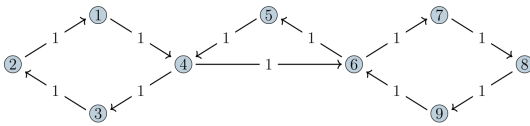
An example of indirect cycles



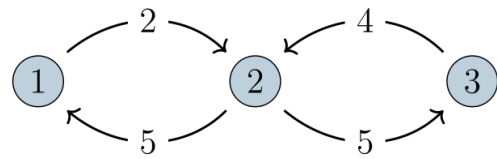
- **Read: equilibrium multiplicity.** The three-bank example shows how cyclic liabilities can support both a good clearing outcome and a bad self-fulfilling default outcome.
- **Read: indirect cycles.** Even if the debt network itself is not cyclic in the relevant way, default costs or fire-sale effects can create an indirect feedback loop through balance sheets.
- **Why it matters.** These figures motivate Proposition 1: multiplicity is about feedback, not simply about large losses.

9.2 Systemic solvency and bailout ordering

Iteratively strongly solvent sets and systemic solvency



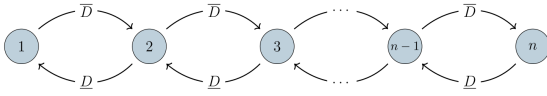
An example where the order of bailouts matters



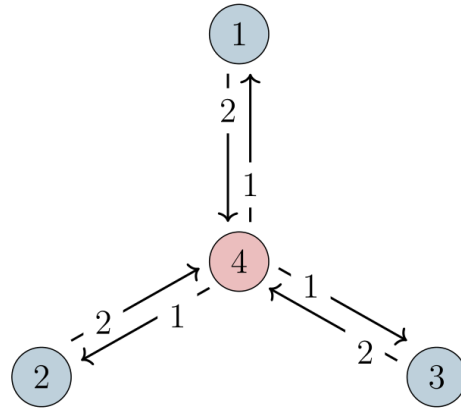
- **Read: systemic solvency.** The left and middle cycles can clear because solvency begins at a strong bank and cascades. The right cycle lacks such a seed and defaults in the worst equilibrium.
- **Read: bailout ordering.** The same set of bailed-out banks can cost different amounts depending on order. Rescuing a debtor first can reduce the cost of rescuing its creditor later.
- **Why it matters.** These visuals explain why the regulator's problem is not just selecting banks; it is selecting a sequence.

9.3 Greedy failure and star-network intuition

An example where the greedy algorithm performs poorly



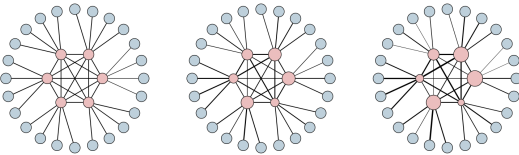
A star network with three peripheral banks



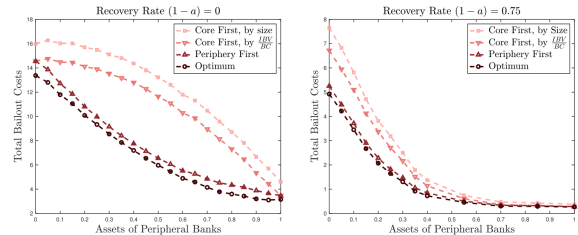
- **Read: greedy failure.** A locally cheap bailout can be globally expensive when repayment cascades run in only one direction.
- **Read: star network.** The center lies on all cycles, but rescuing peripheral debtors can be cheaper because their outside assets are leveraged into payments to the center.
- **Why it matters.** These figures show both sides of the paper's policy message: simple greedy rules can fail, but network structure can still provide sharp guidance.

9.4 Core-periphery simulations

Sample core-periphery networks with different asymmetry

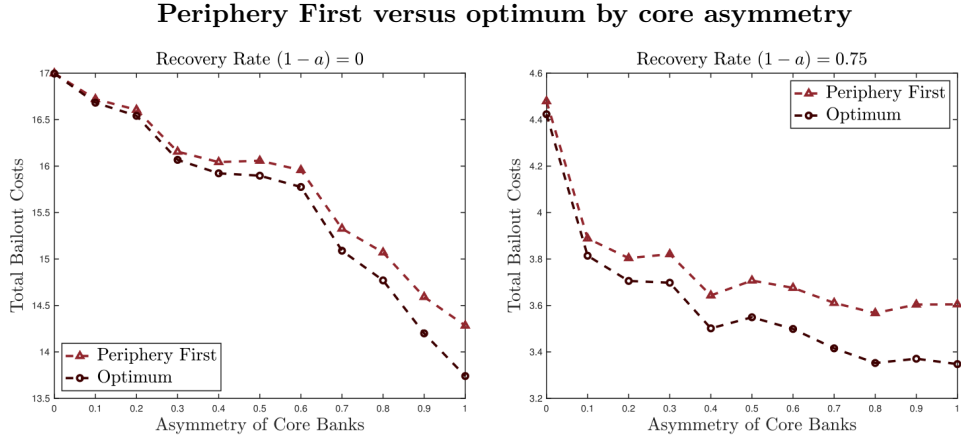


Bailout policies by size of peripheral banks



- **Read: sample networks.** Increasing α makes core-bank exposures and outside assets less symmetric.
- **Read: peripheral size.** Periphery First is most valuable when peripheral banks have intermediate outside assets: enough to leverage, but not enough to be automatically solvent.
- **Why it matters.** These simulations show that the periphery-first intuition survives outside the clean star-network case.

9.5 Asymmetry and the gap to the optimum



- **Read:** When core banks are symmetric, ordering core-bank rescues is relatively unimportant and Periphery First performs close to optimally. As asymmetry rises, the optimal policy exploits detailed differences across core banks, and the gap widens.
- **Why it matters.** The figure separates the paper's robust insight from the hard part. Starting with the periphery is often right; finding the optimal order inside a heterogeneous core remains difficult.

10 Short summary

- **Model:** banks are linked by debt claims; default creates failure costs; bank values solve a fixed point.
- **Multiplicity:** cycles of liabilities can create self-fulfilling defaults and credit freezes.
- **Systemic solvency:** best-equilibrium solvency equals weak balance; worst-equilibrium solvency also requires an iteratively strongly solvent set hitting every cycle.
- **Bailouts:** best-equilibrium bailouts are net imbalances; preventing bad-equilibrium freezes is strongly NP-hard because bailout order matters.
- **Policy lesson:** use indirect bailouts. In star and core-periphery settings, rescuing peripheral debtors first can be cheaper than rescuing the core directly.

- **Simulation lesson:** Periphery First outperforms naive Core First policies, especially when peripheral banks have intermediate assets and when the network permits useful repayment cascades.

11 Conclusion

Core conclusion. Financial networks can generate multiple equilibrium outcomes even for the same balance sheets. A bad equilibrium can look like a credit freeze: banks do not pay because they expect others not to pay, and the expectation is self-confirming.

Economic intuition. A cycle of obligations is both a source of fragility and a channel for rescue. The same network that transmits defaults can transmit solvency if the regulator injects capital at the right nodes.

Takeaway. The cheapest bailout is often not the most visible bailout. Rather than rescuing the largest or most central bank first, a regulator may reduce systemic bailout costs by rescuing smaller peripheral debtors whose payments unlock solvency elsewhere in the network.